

DNS Slave Server Setup Guide

== MASTER SERVER CHANGES (RHEL 9 - 192.168.0.100)

1. Pre-Implementation Checks

Before starting, verify the change request status and ensure you have approval. Login the Master Server via SSH or Console.

****Take a Screenshot: ****

Always create screenshot of configuration files before making changes.

2. Network Configuration

Set the network adapter to Bridged (automatic) and configure the static IP.

****Configure IP Address: ** & **Verify Network: ****

#ifconfig

nmcli con mod ens160 ipv4.addresses "192.168.0.100/24" ipv4.gateway "192.168.0.1"

```
[root@master ~]# nmcli con mod ens160 ipv4.addresses "192.168.0.100/24" ipv4.gateway "192.168.0.1"
[root@master ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.100 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::20c:29ff:fe48:7dae prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:48:7d:ae txqueuelen 1000 (Ethernet)
    RX packets 2639 bytes 652866 (637.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 273 bytes 24486 (23.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 33 bytes 3639 (3.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 33 bytes 3639 (3.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

***Note:** Ensure you bring the connection up or restart network services if the IP doesn't apply immediately. *

```
[root@master ~]# nmcli connection up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
```

****Set Hostname: ****

hostnamectl set-hostname master.radical.com && bash

```
[root@master ~]# hostnamectl set-hostname master.radical.com && bash
[root@master ~]# hostnamectl
  Static hostname: master.radical.com
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 250a08b7dd4642dfab04a7d1882a009d
            Boot ID: d6711112ef9542388560e23b11552216
    Virtualization: vmware
Operating System: Red Hat Enterprise Linux 9.7 (Plow)
      CPE OS Name: cpe:/o:redhat:enterprise_linux:9::baseos
        Kernel: Linux 5.14.0-611.5.1.el9_7.x86_64
    Architecture: x86-64
    Hardware Vendor: VMware, Inc.
    Hardware Model: VMware20,1
Firmware Version: VMW201.00V.24006586.B64.2406042154
```

****Check Connectivity: ****

ping 192.168.0.100 -c 3

```
[root@master ~]# ping 192.168.0.100 -c 3
PING 192.168.0.100 (192.168.0.100) 56(84) bytes of data:
 64 bytes from 192.168.0.100: icmp_seq=1 ttl=64 time=0.049 ms
 64 bytes from 192.168.0.100: icmp_seq=2 ttl=64 time=0.292 ms
 64 bytes from 192.168.0.100: icmp_seq=3 ttl=64 time=0.153 ms

--- 192.168.0.100 ping statistics ---
 3 packets transmitted, 3 received, 0% packet loss, time 2061ms
 rtt min/avg/max/mdev = 0.049/0.164/0.292/0.099 ms
```

3. Repository & Package Installation

****Check Repositories: ****

yum repolist

```
[root@master ~]# yum repolist
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

repo id                                repo name
local-appstream                        RHEL 9 Local - AppStream
local-baseos                           RHEL 9 Local - BaseOS
```

***Note: Ensure valid repositories are enabled (AppStream, BaseOS).**

****Install BIND: ****

yum search bind

```
[root@master ~]# yum search bind
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 22:37:20 ago on Tuesday 02 June 2026 12:21:24 AM.
===== Name & Summary Matched: bind =====
bind.x86_64 : The Berkeley Internet Name Domain (BIND) DNS (Domain Name System)
               : server
```

yum install -y bind.x86_64

```
[root@master ~]# yum install bind.x86_64 -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 22:39:18 ago on Tuesday 02 June 2026 12:21:24 AM.
Package bind-32:9.16.23-31.el9_6.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

4. Configure Master DNS

****Step 1: Update `named.conf`****

Edit the main configuration file to listen on the Master IP and allow queries.

vi /etc/named.conf

```
options {
    listen-on port 53 { 192.168.0.100; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { 192.168.0.0/24; };
    allow-transfer { 192.168.0.101; };
    also-notify { 192.168.0.101; };
}
```

****Step 2: Update Zones File****

Define the zones in the RFC file.

vi /etc/named.rfc1912.zones

```
// Forward Zone
zone "radical.com" IN {
    type master;
    file "fzone.radical.com";
    allow-update { none; };
    allow-transfer { 192.168.0.101; };
    also-notify { 192.168.0.101; };
};

// Reverse Zone
zone "0.168.192.in-addr.arpa" IN {
    type master;
    file "rzone.radical.com";
    allow-update { none; };
    allow-transfer { 192.168.0.101; };
    also-notify { 192.168.0.101; };
};
```

****Step 3: Update Forward Zone****

Add the NS record for the Slave server in the forward zone file.

vi /var/named/fzone.radical.com

```
$TTL 1D
@      IN SOA  master.radical.com. root.radical.com. (
                                                0      ; serial
                                                1D      ; refresh
                                                1H      ; retry
                                                1W      ; expire
                                                3H )    ; minimum

@      NS     master.radical.com.
@      NS     slave.radical.com.
master A      192.168.0.100
slave  A      192.168.0.101
```

****Step 4: Update Reverse Zone****

Add the PTR/NS record for the Slave server in the reverse zone file.

vi /var/named/rzone.radical.com

```
$TTL 1D
@      IN SOA  master.radical.com. root.radical.com. (
                                                0      ; serial
                                                1D      ; refresh
                                                1H      ; retry
                                                1W      ; expire
                                                3H )    ; minimum

@      NS     master.radical.com.
@      NS     slave.radical.com.
100    PTR     master.radical.com.
101    PTR     slave.radical.com.
```

Step5. Validation & Restart

****Check Syntax: ****

```
# named-checkconf /etc/named.conf
```

```
[root@master ~]# named-checkconf /etc/named.conf
[root@master ~]# █
```

*(No output means configuration is valid) *

****Validate Zones: ****

```
# named-checkzone radical.com /var/named/fzone.radical.com
```

```
[root@master ~]# named-checkzone radical.com /var/named/fzone.radical.com
zone radical.com/IN: loaded serial 0
OK
[root@master ~]# █
```

```
# named-checkzone 0.168.192.in-addr.arpa /var/named/rzone.radical.com
```

```
[root@master ~]# named-checkzone 0.168.192.in-addr.arpa /var/named/rzone.radical.com
zone 0.168.192.in-addr.arpa/IN: loaded serial 0
OK
-
```

****Restart Service: ****

```
# systemctl restart named.service
```

```
# systemctl status named.service
```

```
[root@master ~]# systemctl restart named.service
[root@master ~]# systemctl status named.service
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabl>
   Active: active (running) since Tue 2026-06-02 23:15:08 IST; 9s ago
     Process: 3281 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes">
     Process: 3284 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code>
   Main PID: 3285 (named)
      Tasks: 8 (limit: 22780)
     Memory: 25.4M (peak: 25.9M)
        CPU: 177ms
    CGroup: /system.slice/named.service
            └─3285 /usr/sbin/named -u named -c /etc/named.conf
```

6. Firewall & SELinux Configuration

****Allow DNS Traffic: ****

```
# firewall-cmd --permanent --add-service=dns
```

```
# firewall-cmd --permanent --add-port=53/tcp
```

```
# firewall-cmd --permanent --add-port=53/udp
```

```
# firewall-cmd --reload
```

```
# firewall-cmd --list-services
```

```
**Check SELinux: **
```

```
# getenforce
```

```
# setenforce 0
```

(Setting to Permissive temporarily helps troubleshoot transfer issues, Enforcing is recommended for production with proper contexts).

```
**Check Contexts: **
```

```
# ls -lZ /var/named/slaves/
```

```
[root@master ~]# ls -lZ /var/named/slaves/  
total 0
```

The total 0 output on your master server is completely normal.

=====

== SLAVE SERVER SETUP (RHEL 8 - 192.168.0.101)

1. Hostname Configuration

```
**Set Hostname: **
```

```
# hostnamectl set-hostname slave.radical.com
```

```
# hostnamectl
```

```
[root@slave ~]# hostnamectl set-hostname slave.radical.com  
[root@slave ~]# hostnamectl  
  Static hostname: slave.radical.com  
        Icon name: computer-vm  
        Chassis: vm  
        Machine ID: 4dca7b7099d647be8c417b22f74bc687  
        Boot ID: 71638a52afb44b418ef03fb717864da0  
        Virtualization: vmware  
        Operating System: Red Hat Enterprise Linux 8.8 (Ootpa)  
        CPE OS Name: cpe:/o:redhat:enterprise_linux:8::baseos  
        Kernel: Linux 4.18.0-477.10.1.el8_8.x86_64  
        Architecture: x86-64  
[root@slave ~]# █
```

2. Network Verification

```
**Check IP: **
```

ifconfig

```
[root@slave ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.101 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::20c:29ff:fe19:ac45 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:19:ac:45 txqueuelen 1000 (Ethernet)
    RX packets 83 bytes 8329 (8.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 26 bytes 3294 (3.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

*Ensure IP is `192.168.0.101`. *

**Connectivity Check: **

Ping the Master server to ensure layer 3 connectivity.

ping 192.168.0.100 -c 4

```
[root@slave ~]# ping 192.168.0.100 -c 4
PING 192.168.0.100 (192.168.0.100) 56(84) bytes of data.
64 bytes from 192.168.0.100: icmp_seq=1 ttl=64 time=0.683 ms
64 bytes from 192.168.0.100: icmp_seq=2 ttl=64 time=1.08 ms
64 bytes from 192.168.0.100: icmp_seq=3 ttl=64 time=0.588 ms
64 bytes from 192.168.0.100: icmp_seq=4 ttl=64 time=2.20 ms

--- 192.168.0.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.588/1.139/2.204/0.642 ms
[root@slave ~]#
```

3. Hosts File Configuration

Update the hosts file for local name resolution before DNS is fully active.

vi /etc/hosts

```
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.0.100 master.radical.com master
192.168.0.101 slave.radical.com slave
```

**Verify: **

cat /etc/hosts

```
[root@slave ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.0.100 master.radical.com master
192.168.0.101 slave.radical.com slave
```

4. Install BIND on Slave

**Check Existing Packages: **

rpm -qa | grep bind

```
[root@slave ~]# rpm -qa | grep bind
rpcbind-1.2.5-10.el8.x86_64
bind-9.11.36-8.el8.x86_64
keybinder3-0.3.2-4.el8.x86_64
bind-libs-9.11.36-8.el8.x86_64
bind-utils-9.11.36-8.el8.x86_64
bind-license-9.11.36-8.el8.noarch
bind-libs-lite-9.11.36-8.el8.x86_64
python3-bind-9.11.36-8.el8.noarch
[root@slave ~]#
```

****Mount Media (If local repo needed): ****

mount /dev/cdrom /mnt

```
[root@slave ~]# mount /dev/cdrom /mnt
mount: /mnt: WARNING: device write-protected, mounted read-only.
[root@slave ~]#
```

****List and Install: ****

yum list bind*

yum install -y bind.x86_64

```
[root@slave ~]# yum install -y bind.x86_64
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 5:36:05 ago on Tuesday 02 June 2026 08:23:53 AM EDT.
Package bind-32:9.11.36-8.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

****Verify Installation: ****

rpm -qa | grep bind

```
[root@slave ~]# rpm -qa | grep bind
rpcbind-1.2.5-10.el8.x86_64
bind-9.11.36-8.el8.x86_64
keybinder3-0.3.2-4.el8.x86_64
bind-libs-9.11.36-8.el8.x86_64
bind-utils-9.11.36-8.el8.x86_64
bind-license-9.11.36-8.el8.noarch
bind-libs-lite-9.11.36-8.el8.x86_64
python3-bind-9.11.36-8.el8.noarch
[root@slave ~]#
```

5. Configuration Backup

****List Config Files: ****

rpm -qc bind.x86_64


```
[root@slave ~]# rpm -qc bind.x86_64
/etc/logrotate.d/named
/etc/named.conf
/etc/named.rfc1912.zones
/etc/named.root.key
/etc/rndc.conf
/etc/rndc.key
/etc/sysconfig/named
/var/named/named.ca
/var/named/named.empty
/var/named/named.localhost
/var/named/named.loopback
[root@slave ~]#
```

****Backup Original Config: ****

```
# cp /etc/named.conf /tmp/named.conf.bkp1
```

6. Configure `named.conf` on Slave

****Edit Configuration: ****

```
# vi /etc/named.conf
```

```
options {
    listen-on port 53 { 192.168.0.101; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     {192.168.0.0/24; };
    allow-transfer  { none; };
}
```

****Validate Syntax: ****

```
# named-checkconf /etc/named.conf
```

```
[root@slave ~]# named-checkconf /etc/named.conf
[root@slave ~]#
```

7. Configure Slave Zones

****Backup Zones File: ****

```
# cp /etc/named.rfc1912.zones /tmp/named.rfc1912.zones.bkp1
```

```
[root@slave ~]# cp /etc/named.rfc1912.zones /tmp/named.rfc1912.zones.bkp1
cp: overwrite '/tmp/named.rfc1912.zones.bkp1'? yes
```

****Edit Zones: ****

```
# vi /etc/named.rfc1912.zones
```

```
#####
// Forward Zone - SLAVE
zone "radical.com" IN {
    type slave;
    file "slaves/fzone.radical.com";
    masters { 192.168.0.100; };
};

// Reverse Zone - SLAVE
zone "0.168.192.in-addr.arpa" IN {
    type slave;
    file "slaves/rzone.radical.com";
    masters { 192.168.0.100; };
};
```

8. Directory Permissions

The `named` user must have write permissions to store transferred zone files.

```
# cd /var/named/
```

```
# ll
```

```
[root@slave ~]# cd /var/named/
[root@slave named]# ll
total 16
drwxrwx---. 2 named named 23 Jun 2 07:05 data
drwxrwx---. 2 named named 60 Jun 2 13:50 dynamic
-rw-r-----. 1 root named 2253 Feb 8 2023 named.ca
-rw-r-----. 1 root named 152 Feb 8 2023 named.empty
-rw-r-----. 1 root named 152 Feb 8 2023 named.localhost
-rw-r-----. 1 root named 168 Feb 8 2023 named.loopback
drwxrwx---. 2 named named 56 Jun 2 08:41 slaves
[root@slave named]#
```

****Set Ownership: ****

```
# chown -R named:named /var/named/slaves/
```

```
# chmod 770 /var/named/slaves/
```

```
# ll -ld /var/named/slaves/
```

```
[root@slave named]# chown -R named:named /var/named/slaves/
[root@slave named]# chmod 770 /var/named/slaves/
[root@slave named]# ll -ld /var/named/slaves/
drwxrwx---. 2 named named 56 Jun 2 08:41 /var/named/slaves/
[root@slave named]#
```

9. Enable and Start Service

```
# systemctl enable --now named.service
```

```
# systemctl status named.service
```

```
[root@slave named]# systemctl enable --now named.service
[root@slave named]# systemctl status named.service
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2026-06-02 12:50:04 EDT; 1h 19min ago
     Main PID: 1095 (named)
        Tasks: 5 (limit: 22897)
       Memory: 18.8M
      CGroup: /system.slice/named.service
              └─1095 /usr/sbin/named -u named -c /etc/named.conf
```

10. Verify Zone Transfer

```
# ll /var/named/slaves/
```

```
[root@slave named]# ll /var/named/slaves/
total 8
-rw-r--r--. 1 named named 284 Jun  2 08:39 fzone.radical.com
-rw-r--r--. 1 named named 354 Jun  2 08:39 rzone.radical.com
```

Note: Check if the zone files have been automatically copied from Master to Slave.

11. Firewall & SELinux on Slave

****Configure Firewall: ****

```
# firewall-cmd --permanent --add-service=dns
```

```
# firewall-cmd --permanent --add-port=53/tcp
```

```
# firewall-cmd --permanent --add-port=53/udp
```

```
# firewall-cmd --reload
```

```
# firewall-cmd --list-services
```

****SELinux Check: ****

```
# getenforce
```

```
#setenforce 0
```

****Restore Contexts (If needed): ****

```
# restorecon -Rv /var/named/slaves/
```

```
[root@slave named]# restorecon -Rv /var/named/slaves/
[root@slave named]#
```

resets the SELinux security contexts of the slaves directory to their default values.

****Verify Listening Ports: ****

```
# netstat -antlpu | grep named
```

```
[root@slave named]# netstat -antlpu | grep named
tcp        0      0 192.168.0.101:53      0.0.0.0:*              LISTEN      1095/named
tcp        0      0 127.0.0.1:953         0.0.0.0:*              LISTEN      1095/named
tcp6       0      0 :::1:53               :::*                    LISTEN      1095/named
tcp6       0      0 :::1:953              :::*                    LISTEN      1095/named
udp        0      0 192.168.0.101:53      0.0.0.0:*              1095/named
udp6       0      0 :::1:53               :::*                    1095/named

[root@slave named]#
```

=====

Final Verification Phase

Test the DNS resolution from both servers to ensure the Master-Slave setup is working correctly.

Step From the Slave Server (192.168.0.101)

Test if the Slave can resolve records it learned from the Master.

****Forward Lookup: ****

nslookup master.radical.com 192.168.0.101

```
[root@slave named]# nslookup master.radical.com 192.168.0.101
Server:      192.168.0.101
Address:     192.168.0.101#53

Name:   master.radical.com
Address: 192.168.0.100

[root@slave named]#
```

nslookup slave.radical.com 192.168.0.101

```
[root@slave named]# nslookup slave.radical.com 192.168.0.101
Server:      192.168.0.101
Address:     192.168.0.101#53

Name:   slave.radical.com
Address: 192.168.0.101
```

****Reverse Lookup: ****

nslookup 192.168.0.100 192.168.0.101

```
[root@slave named]# nslookup 192.168.0.100 192.168.0.101
100.0.168.192.in-addr.arpa      name = master.radical.com.
```

Step From the Master Server (192.168.0.100)

Verify that the Master acknowledges the Slave.

```
# nslookup slave.radical.com 192.168.0.100
```

```
[root@master ~]# nslookup slave.radical.com 192.168.0.100
Server:      192.168.0.100
Address:     192.168.0.100#53
```

```
Name:   slave.radical.com
Address: 192.168.0.101
```

=====

Conclusion

Setup Status: Successfully Completed

The DNS Slave server (**192.168.0.101**) has been successfully deployed and configured. Zone transfers from the Master (**192.168.0.100**) to the Slave are functioning correctly. Both servers are fully operational, providing DNS redundancy for the **radical.com** domain.

Status: Verified